

Adam Rust

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EDUCATION

University of Wisconsin-Madison | Madison, WI

May 2027

Bachelor of Science: Computer Engineering

- GPA: 3.408 Cumulative.
- Awards: *Dean's List (2 Semesters)*, *Deans Honor List (1 Semester)*.
- Relevant Coursework: Signals Processing, Circuit Analysis, Digital Systems, Electrodynamics, Microprocessor Systems, Machine Organization.

TECHNICAL PROJECTS

FPGA-Based Hardware Control

Aug 2025 – Dec 2025

- Implemented both Moore and Mealy FSMs in SystemVerilog to coordinate internal control signals and sequential state transitions.
- Developed a 7-segment decoder using combinational logic to translate 4-bit truth tables into a visual numeric display.
- Verified design functionality through comprehensive RTL simulation in QuestaSim prior to hardware deployment.

Microcontroller-Controlled LED PCB Game.

Jan 2026 – May 2026

- Routed a circular cascaded array of 12 addressable RGB LEDs, implementing strict power decoupling.
- Engineered a dual-rail power system to safely drop a 5V power source to a stable 3.3V logic environment.
- Implemented a hardware-level voltage supervisor and watchdog timer to prevent software and hardware lockups.

Standalone Custom 8-Bit Digital Calculator

May 2026 – Present

- Developed schematics and a 2-layer PCB layout using a direct-pin I/O keypad array, an ATtiny-1617 microcontroller, and a dedicated SPI 7-segment display.
- Integrated test points across the board layout to allow for real-time signal debugging and voltage validation.

Embedded RTOS Application

Jan 2026 – May 2026

- Developed a multi-task embedded application using FreeRTOS on an ARM Cortex-M processor, managing race conditions via queues, semaphores, and event groups.
- Implemented serial communication protocols (UART, I2C, SPI) to interface with external peripherals.
- Designed interrupt-driven I/O routines using NVIC priority configuration and ISRs utilizing volatile global variables.

Low-Level Software Architecture x86-64 Assembly

Aug 2024 – May 2025

- Optimized data manipulation using various bitwise operations, logical shifts, and bit-masking to parse raw 32-bit integers into formatted IPv4 addresses.
- Implemented low-level multiplayer tic-tac-toe program utilizing CPU status flags and conditional branching to manage game-loop control flow.

TECHNICAL SKILLS

Languages: SystemVerilog (HDL), C/C++, Java, Python, Bash, Assembly, HTML/CSS.

Tools: QuestaSim, Quartus, MATLAB, LTspice, Make, Git/GitHub, Altium Designer, FreeRTOS.

Hardware & Lab: PCB Design, Mixed-Signal Routing, Oscilloscope, Digital Multimeter, Soldering.

INVOLVEMENT / LEADERSHIP

University of Wisconsin Marching Band | UW-Madison | Madison, WI

Aug 2023 – Present

Saxophone Rank Leader

- Selected as Rank Leader to organize and prepare the saxophone section for high-profile field performances.
- Directed marching drills and mentored individual members on technique and field awareness.
- Built a teamwork-oriented environment by adapting my leadership style to better suit different communication and learning styles.